AHMAD HASSAN — FULL-STACK CASE STUDIES

Technical Overview, System Architecture & Client Solutions

Portfolio Showcase Document | Contact: ahmad.hassan.dev@gmail.com | github.com/Ahmad-Haassan

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CASE STUDY 1: ShopLens (E-Commerce Intelligence Platform)

Problem Statement: Online shoppers waste significant time cross-referencing prices, delivery fees, and genuine reviews across multiple e-commerce stores with inconsistent formats.
Proposed Solution: A unified web platform where users input product links to receive normalized price comparisons, standardized feature specifications, and synthesized customer review summaries.
Architecture & Stack: Next.js 16 App Router on frontend, Python data extraction workers, structured JSON schemas, and Tailwind CSS for rapid responsive rendering.
Key Outcomes: Normalized disparate product attributes into clean UI comparison tables; delivered sub-100ms client-side filtering.

CASE STUDY 2: Flexivo (Desktop Assistant & Workflow Automation)

Problem Statement: Repetitive system actions, hotkeys, and app switching create context friction for daily knowledge workers.
Proposed Solution: An experimental desktop assistant built with speech recognition pipelines, system hotkey listeners, and automated task execution.
Architecture & Stack: Python, PyQt / Desktop UI, Speech Recognition APIs, OS Automation hooks.
Key Outcomes: Enabled hands-free system command routing and routine task execution with minimal CPU overhead.

CASE STUDY 3: Enterprise Webhook & WhatsApp Routing System

Problem Statement: Small businesses struggle to respond promptly to lead inquiries outside standard business hours, causing lost conversion opportunities.
Proposed Solution: An automated messaging relay connecting WhatsApp Business Webhooks to structured routing logic and auto-responders.
Architecture & Stack: Node.js, Express webhook listener, JSON Schema validation, REST API dispatchers.
Key Outcomes: 100% automated first-touch response time; verified webhook signatures preventing unauthorized access.

ENGINEERING STANDARDS & GUARANTEE

- **Clean, Documented Code:** Every project is delivered with modular code, comprehensive README setup instructions, and clean Git history.
- **Performance-First:** 90+ score guarantee on Google Lighthouse for SEO, Accessibility, and Core Web Vitals.
- **Cross-Platform Assurance:** Tested rigorously across Chrome, Safari, Firefox, Edge, and iOS/Android mobile viewports.
- **Free Post-Delivery Bug Support:** Ensuring your deployment stays smooth and trouble-free.